



Graph Learning Pipeline for Architectural Floor Plans

Node classification on the Narkomfin Building using GraphSAGE — from redrawn geometry to a labelled spatial graph and predicted room types.

FINAL ASSIGNMENT

MaCAD 2025/26 · IAAC

Digital Tools for Graph Machine Learning

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What we cover.

01

Modified Swiss Dwellings

A benchmark of 5,372 residential floor plans encoded as spatial graphs

02

Literature

Two papers and one pretrained GraphSAGE-Pool model

03

Data Structure

Nodes, edges, features and ground-truth room-type labels

04

The Narkomfin Building

Dom-kommuna, two unit types, one ideal graph-ML subject

05

Graph Analysis

Centrality measures, shortest paths and spatial logic

06

Node Classification

Room-type prediction from zoning features using GraphSAGE

Modified Swiss Dwellings.

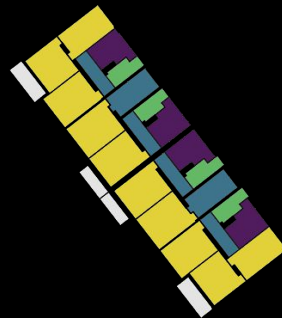
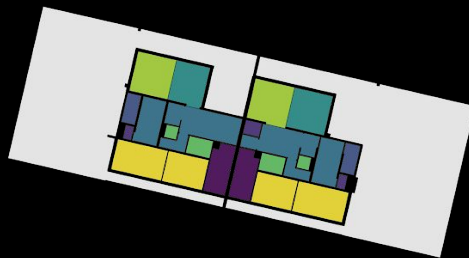
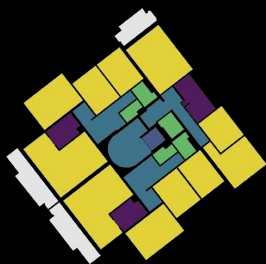
5,372 floor plans

Multi-apartment clusters

Corridor buildings

Single units

Graph modality



Three sample plans — a wide variety of residential typologies, scales and orientations · node colour encodes room type

5,372

Floor Plans

9

Room Types

4

Zone Types

.pickle

Graph Format

Two papers, one pipeline.

PAPER 01

MSD: A Benchmark Dataset for Floor Plan Generation of Building Complexes



ECCV 2024 · TU Delft

- 5,372 Swiss residential floor plans
- 3 modalities: image, geometry, graph
- Graph encodes spatial zones, not room types

Primary data source for this project.

PAPER 02

GNN for Node Classification & Attribute Allocation in Architectural BIM



eCAADe 2024 · Cardiff University

- Same MSD dataset, graphs via TopologicPy
- Benchmarks GCN, GAT, GraphSAGE
- GraphSAGE-Pool, 4 layers, ~95% accuracy

Pretrained model applied to our floor plan.

What the graph encodes.

graph_in — model input

Attribute	Type	Values
zoning_type	int	0 · 1 · 2 · 3
connectivity	str	'door' · 'entrance'

graph_out — ground truth

Attribute	Type	Description
room_type	int	0 – 8
geometry	Polygon	2D room outline
centroid	Point	centre of room

Room types (0 – 8)

ID	Room	ID	Room
0	Bedroom	5	Stairs
1	Living room	6	Storeroom
2	Kitchen	7	Bathroom
3	Dining	8	Balcony
4	Corridor		

Zone types (0 – 3)

ID	Zone	Typical rooms
0	Private	bedroom
1	Dynamic	corridor, entrance
2	Static	bedroom, bathroom
3	Functional	kitchen, storeroom

From floor plan to graph.

Zone label per node

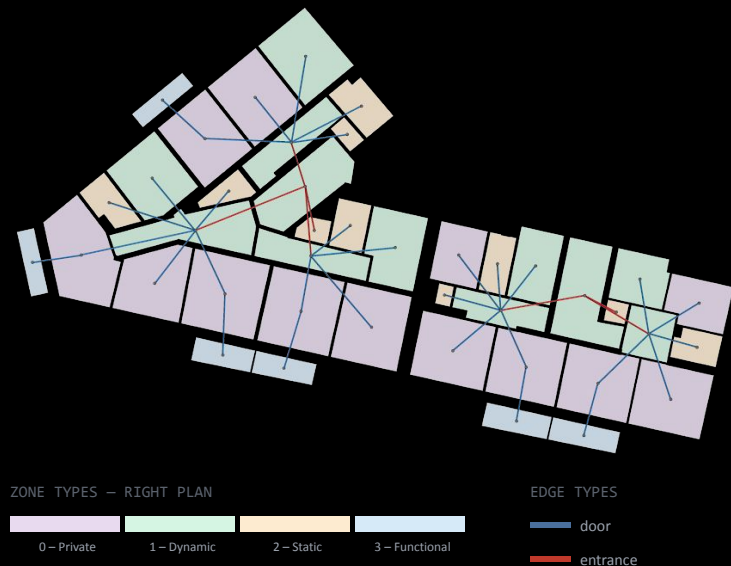
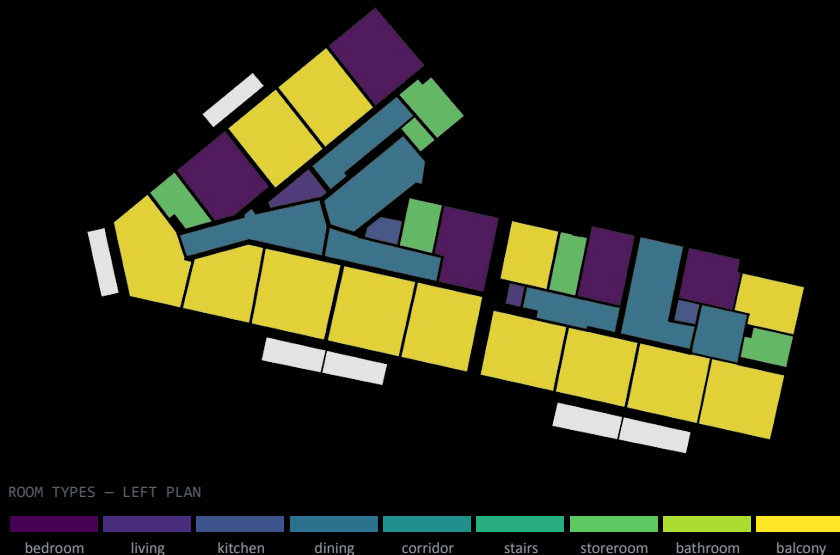
zoning_type input

Graph topology

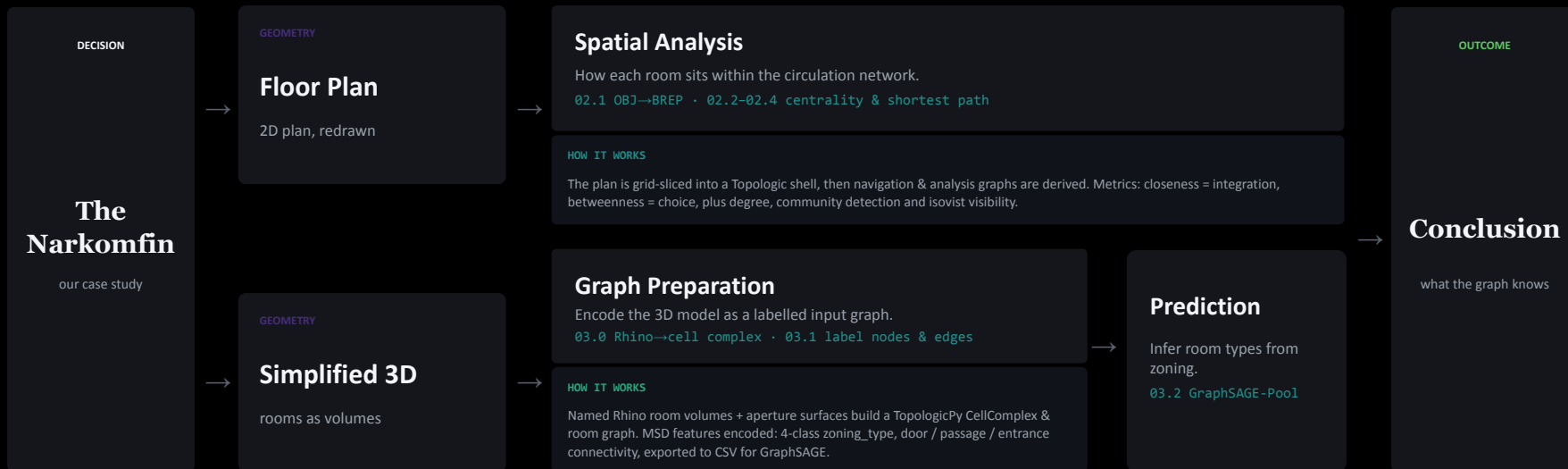
door / entrance edges

Room type prediction

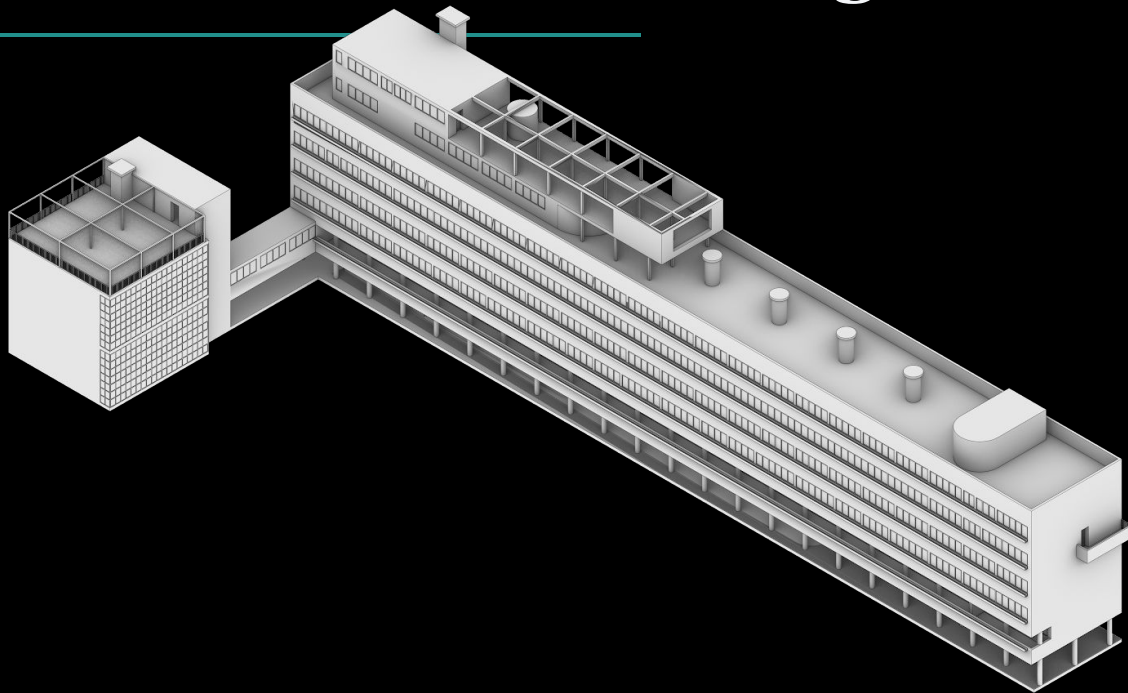
GraphSAGE-Pool output



One building, two paths, one read.



A machine for collective living.



ARCHITECTS

Ginzburg & Milinis

YEAR

1930

LOCATION

Moscow, Russia

TPOLOGY

Dom-kommuna

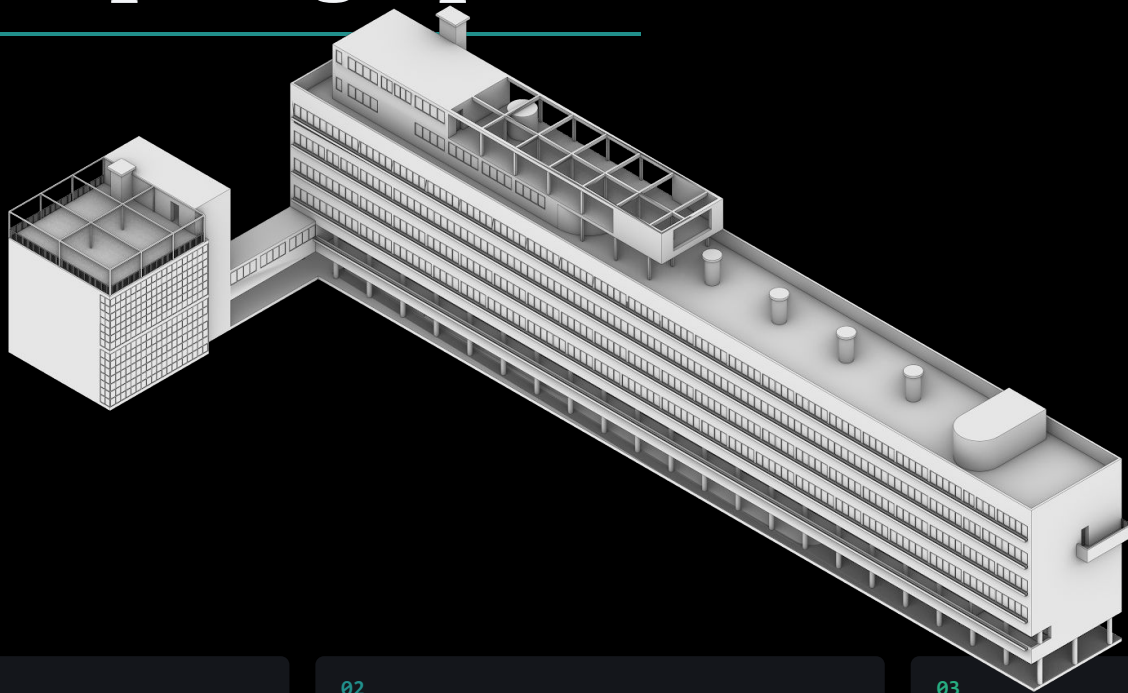
SCALE

54 units · 6 floors

UNIT TYPES

F · K

Why it maps to graph ML.



01

Clear spatial hierarchy

Discrete living cells and corridors form a readable room-to-room structure.

02

Repetitive unit typologies

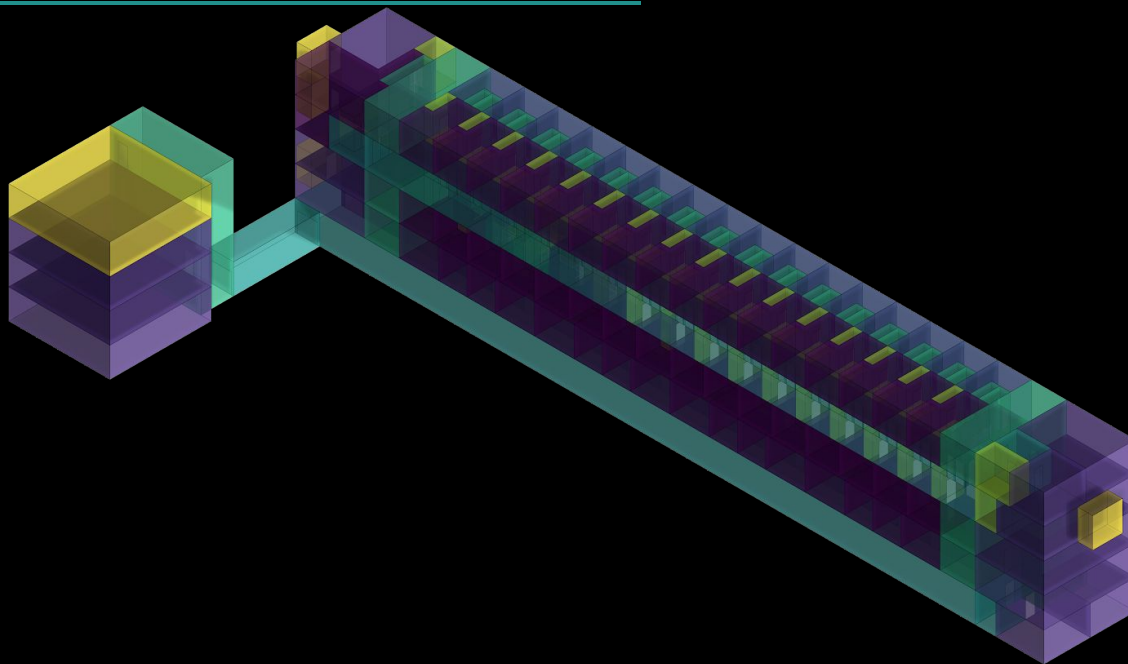
Two unit types — F-type and K-type — repeat cleanly across every floor.

03

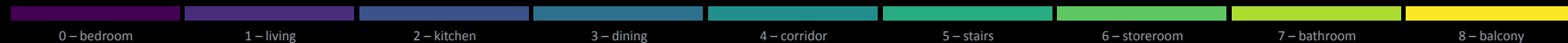
Explicit functional zoning

A dom-kommuna where living, communal and circulation zones are clearly defined.

Simplified model, rooms as volumes.

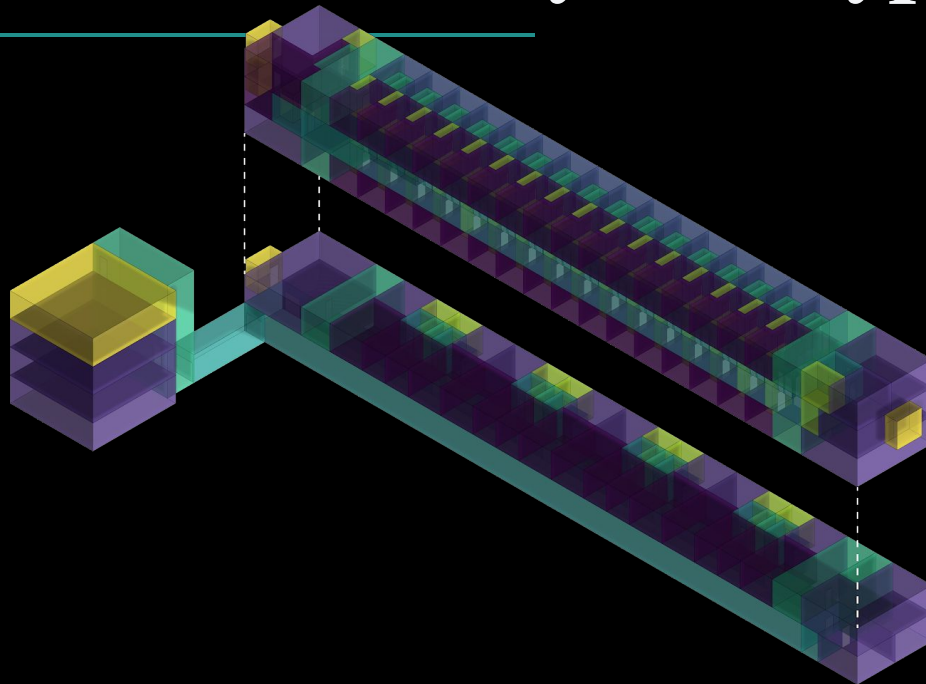


ROOM COLOUR ENCODES ROOM TYPE



Exploded axonometric by floor type.

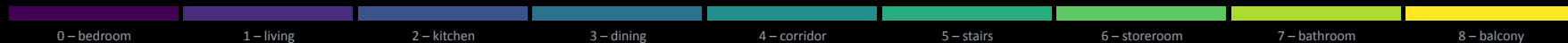
Amenities



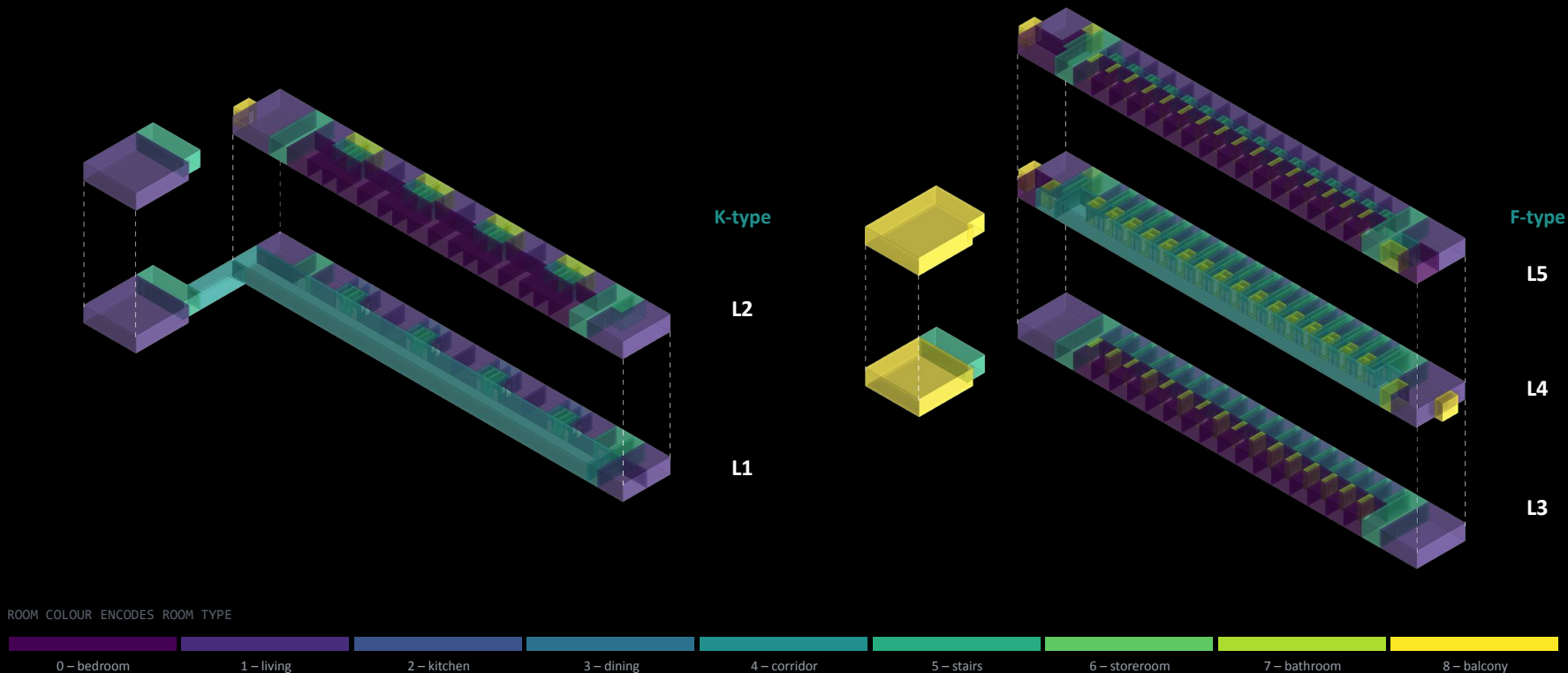
F - type

K - type

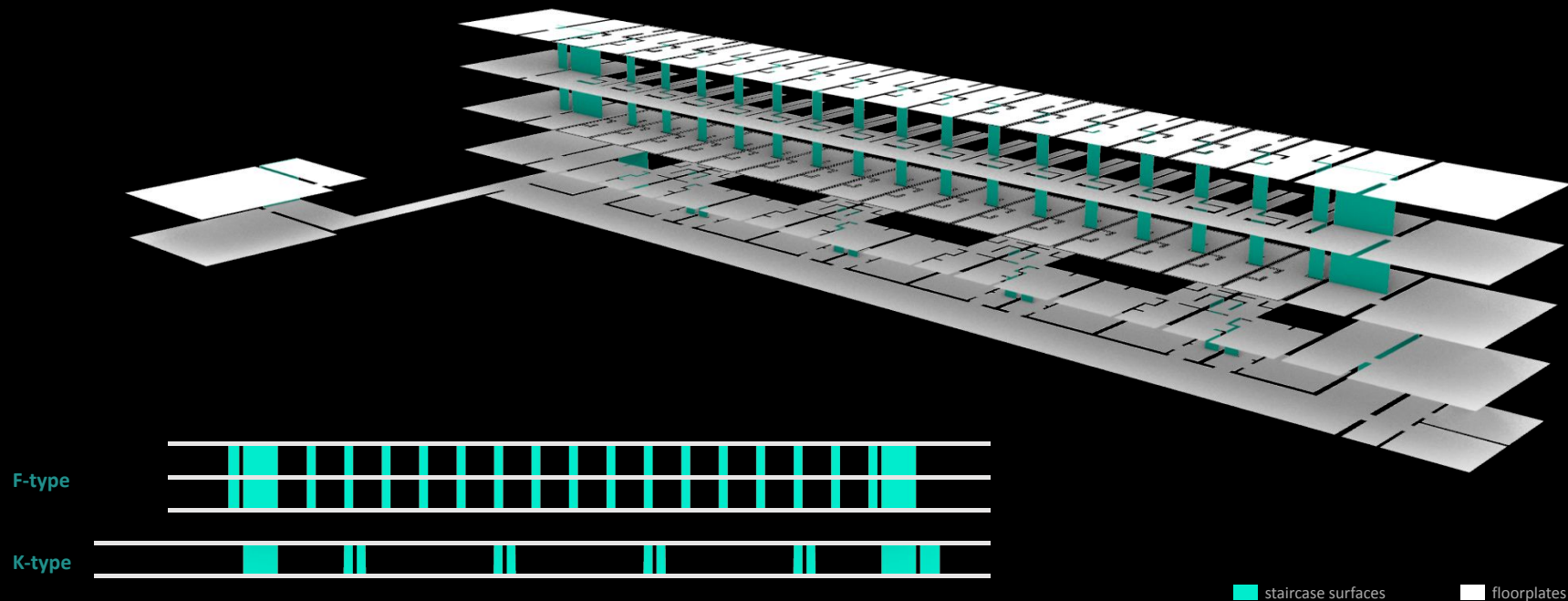
ROOM COLOUR ENCODES ROOM TYPE



K-type & F-type, five levels.



Exported Floorplate



Building graph, Type K.

3,476 nodes

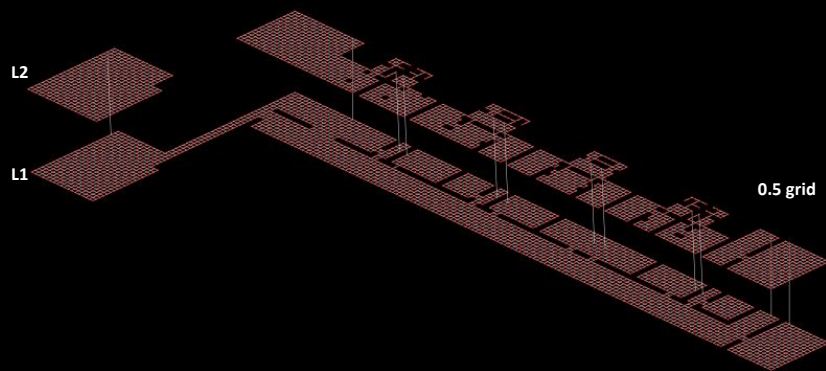
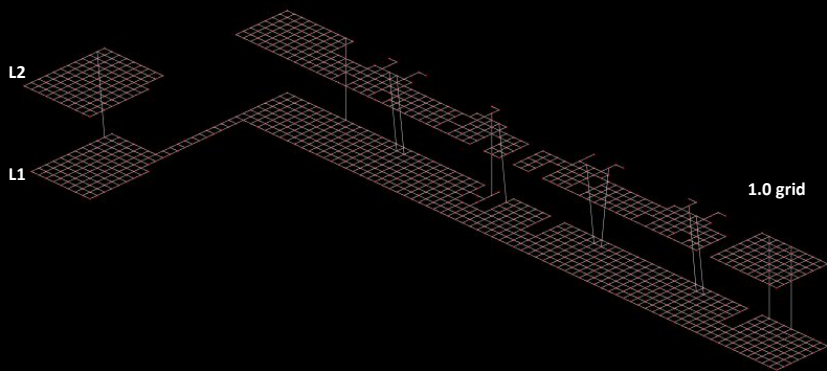
0.5m grid, L1+L2

12 stair edges

cross-floor links

Density 0.0028

linear topology



- L1 is a dense, continuous mesh. The long corridor spine produces an unbroken chain of well-connected cells running the full length of the unit.
- L2 is visibly fragmented. Each room appears as a separate island of connected cells, linked to L1 only through vertical stair edges.
- The stair edges cluster in the centre of the plan, creating a narrow band of cross-floor connectivity. West-end rooms and east-end balconies are most remote.
- Graph density is very low (0.0028), reflecting the linear, corridor-dominated topology. The plan is fundamentally sequential.

Building graph, Type F.

6,269 nodes

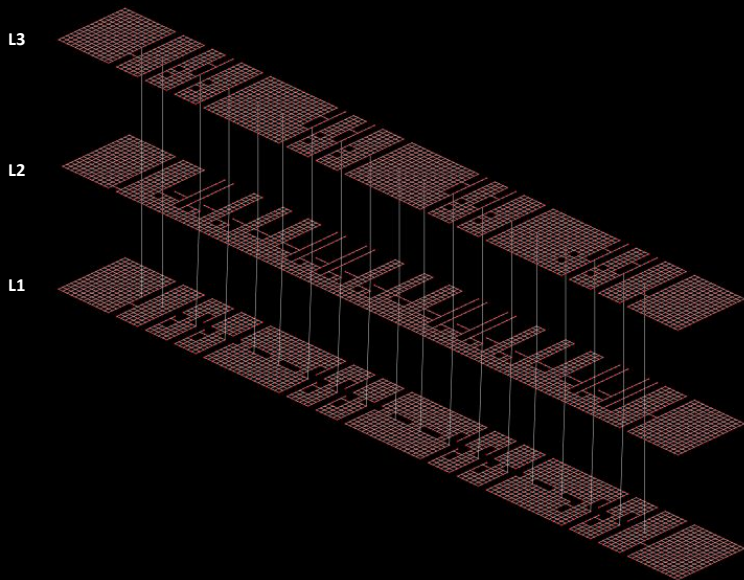
0.5m grid, L1+L2+L3

19 stair edges

cross-floor links

Density 0.000563

linear topology



Key findings

- L2 (middle) is the densest, most continuous mesh. This is the spatial backbone of the Type F.
- L1 (bottom) and L3 (top) are both fragmented into room islands, symmetric about the corridor.
- Each stair produces two vertical connections (L1→L2 and L2→L3), creating a ladder-like structure.
- The fragmentation is symmetric: rooms exist both above and below the corridor floor.

Closeness centrality, Type K.

Corridor = integrator

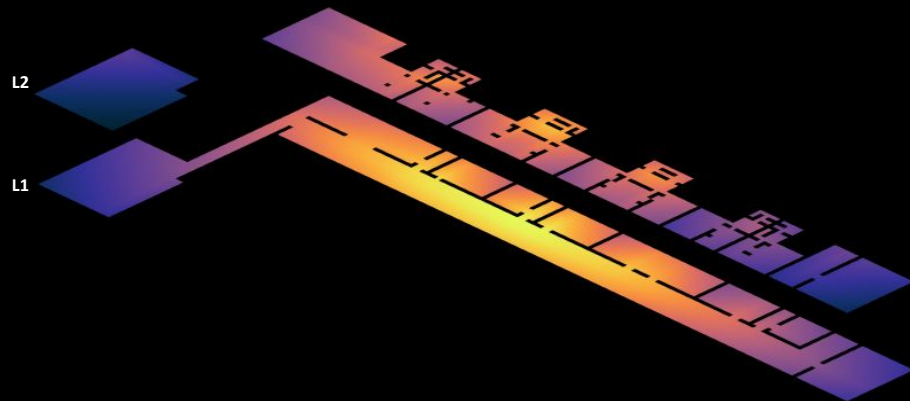
highest closeness on L1 spine

L2 = topologically deep

uniformly low closeness

Privacy gradient upward

L1 public, L2 private



Key findings

- The brightest region forms a hot band along the L1 main corridor, centre-to-east — the topological heart of the apartment with shortest average path distance to every other cell.
- The gradient is strikingly linear. Closeness falls off steadily toward both ends — west toward the kitchen block, east toward the far rooms.
- L2 rooms are uniformly blue-purple, confirming they are topologically deep. The east and west extremes of L2 are the darkest — the most remote spaces.
- The corridor is not merely a passage but the primary spatial integrator. The privacy gradient runs upward from L1 to L2, achieved through topological distance.

high closeness

low closeness

Closeness centrality, Type F.

L2 = topological centre

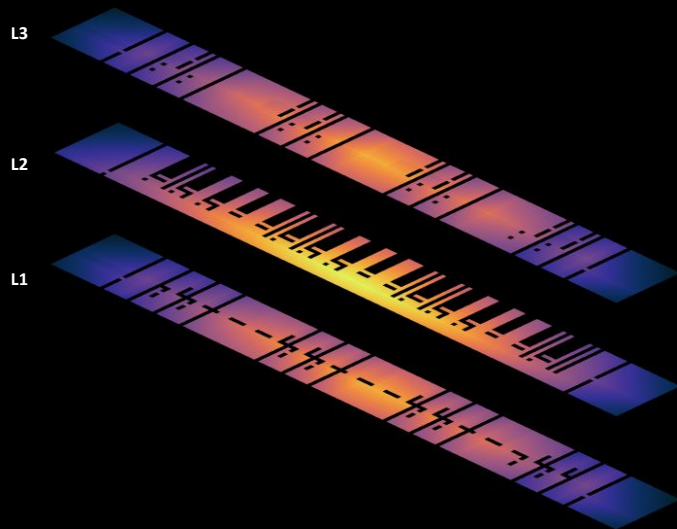
highest closeness on middle floor

Bidirectional gradient

drops equally up and down

Balanced accessibility

L1 and L3 equidistant from corridor



Key findings

- Highest closeness sits on L2 (middle floor), concentrated at the centre of the plan.
- The gradient is bidirectional which means accessibility decreases both downward to L1 and upward to L3.
- L1 and L3 rooms are uniformly blue-purple. It is topologically deep in both directions.
- The middle-floor corridor is equidistant from rooms on both sides, producing a balanced accessibility profile.

high closeness

low closeness

Betweenness centrality, Type K.

Single routing spine

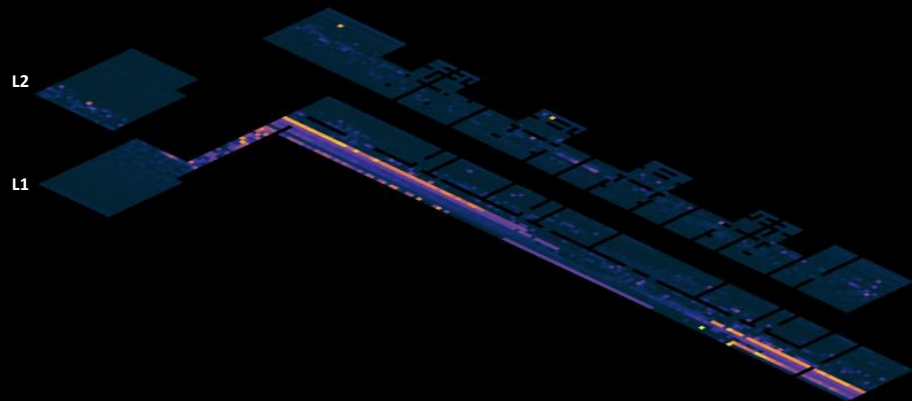
all movement funnels through L1 centreline

Stair landings = chokepoints

localised betweenness spikes

L2 = dead ends

zero through-traffic



Key findings

- The highest betweenness forms a razor-sharp hot line along the L1 corridor centreline. Virtually all cross-building movement passes through a single narrow band of cells.
- Stair landing zones on L1 show localised betweenness spikes — critical chokepoints absorbing all vertical traffic. Blocking any stair would isolate the corresponding L2 rooms.
- L2 rooms are almost entirely dark meaning near-zero betweenness. They are destination spaces only; no shortest path between other cells passes through them.
- The plan concentrates all betweenness into a single linear spine, making the building efficient but fragile. Any corridor obstruction severs circulation.

high betweenness

low betweenness

Betweenness centrality, Type F.

Double-loaded corridor

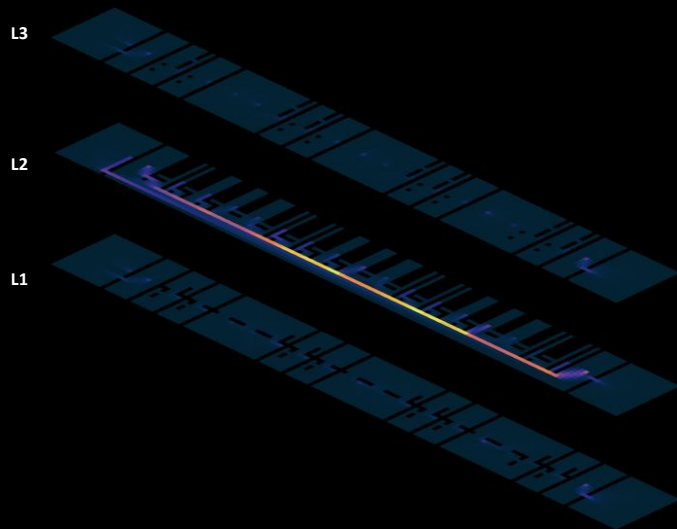
serves rooms above and below

Doubly loaded stairs

absorb traffic in both directions

Higher fragility

obstruction disconnects 2 floors, not 1



Key findings

- An equally sharp hot line on L2 — the corridor carries a double load, serving rooms on two floors.
- Stair landings absorb traffic from both directions (up and down), making them doubly loaded chokepoints.
- L1 and L3 rooms are entirely dark — pure destination spaces with no through-traffic.
- Any L1↔L3 path must traverse L2 as a mandatory intermediary. The corridor is more critical than in Type K.

high betweenness

low betweenness

Shortest path, Type K vs Type F.

K = L-shaped

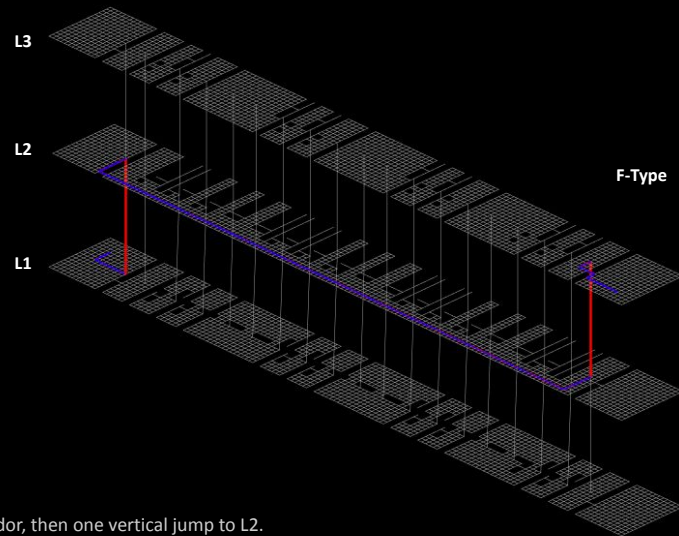
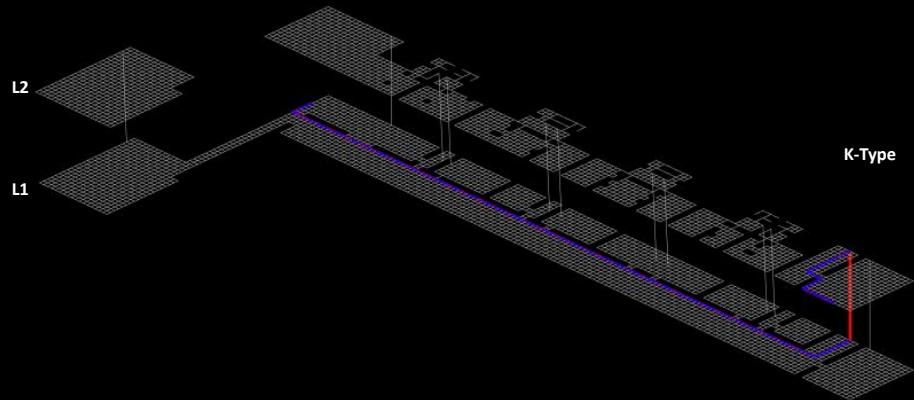
horizontal then one vertical jump

F = Z-shaped

vertical, horizontal, vertical

Corridor dominates

L2 segment is longest portion of F path



■ topological path (red) ■ straightened path (blue)

- Type K: L-shaped path — horizontal along L1 corridor, then one vertical jump to L2.
- Type F: Z-shaped path — vertical up to L2, horizontal along corridor, vertical up to L3.
- Both: straightened (blue) path closely tracks topological (red) — the narrow corridor leaves little room for geometric optimisation.
- The L2 corridor segment dominates the Type F path. Stair placement controls cross-floor routing efficiency in both types.

Degree centrality, Type K vs Type F.

Uniform connectivity

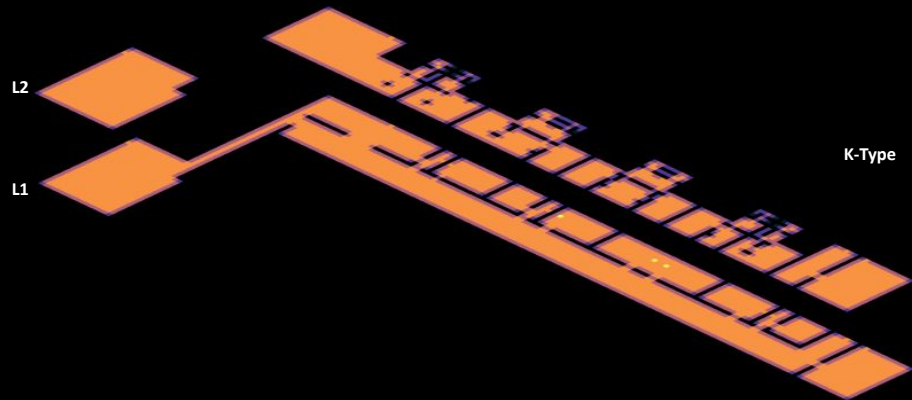
nearly all cells have 4 neighbours

Stair cells = outliers

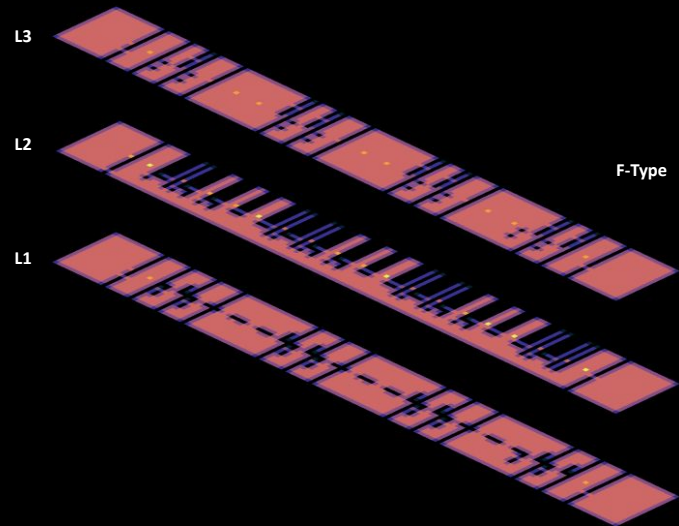
only nodes with degree 5 or 6

Flat local topology

hierarchy only visible at global scale



K-Type



F-Type



- Both types show nearly uniform degree — the plan is topologically flat at the local scale.
- Yellow spots at stair landings where vertical edges add connections (degree 5 in K, up to 6 in F).
- Room interiors have the same local connectivity as corridor interiors. The hierarchy only emerges at the global scale.

Clustering coefficient, Type K vs Type F.

All zeros

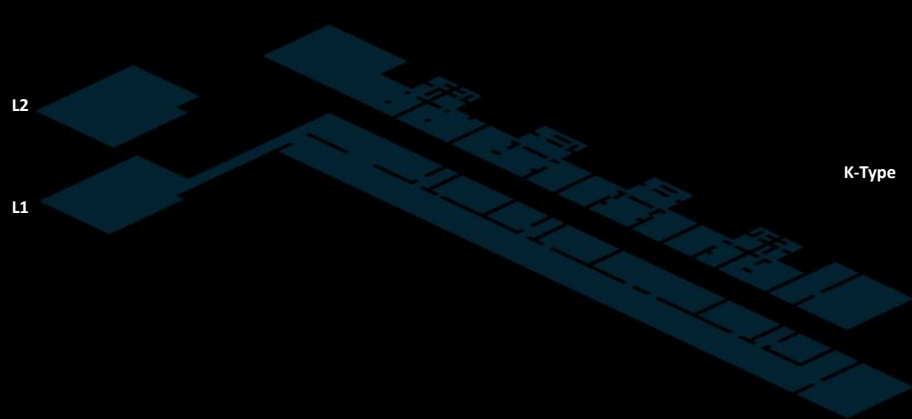
4-neighbour grid cannot form triangles

No loops

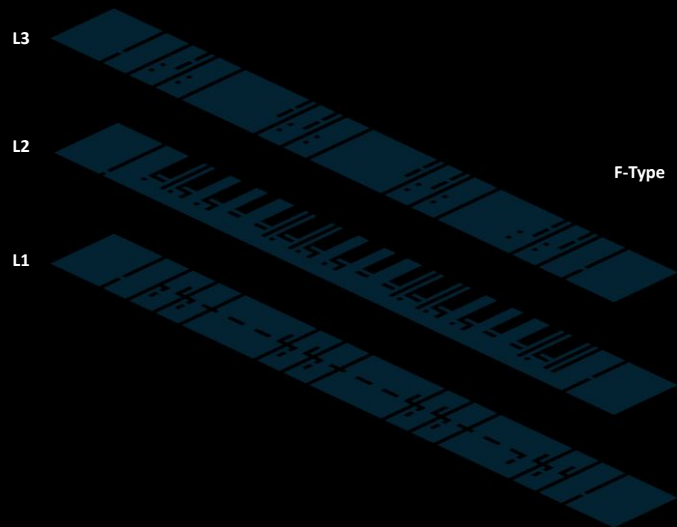
no ring corridors or courtyards

Sequential logic

forward or back, no shortcuts



K-Type



F-Type



- Both: uniformly zero. In a 4-neighbour rectilinear grid, no two neighbours of a cell are themselves neighbours — triangles cannot form.
- Neither apartment has ring corridors, courtyards, or loop connections. Both enforce strictly sequential spatial logic — you can only go forward or back.

Community detection, Type K.

Vertical slices

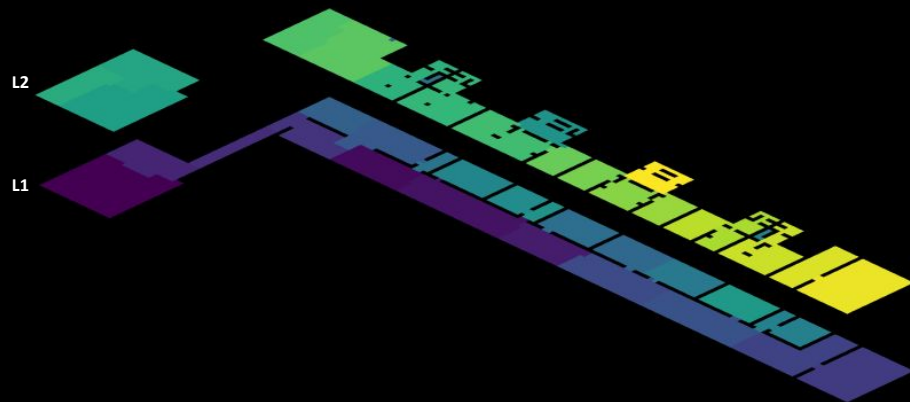
not horizontal floor layers

Stairs bind rooms to corridor

L2 rooms join nearest L1 segment

Longitudinal bands

communities cut perpendicular to corridor



ROOM TYPES



Key findings

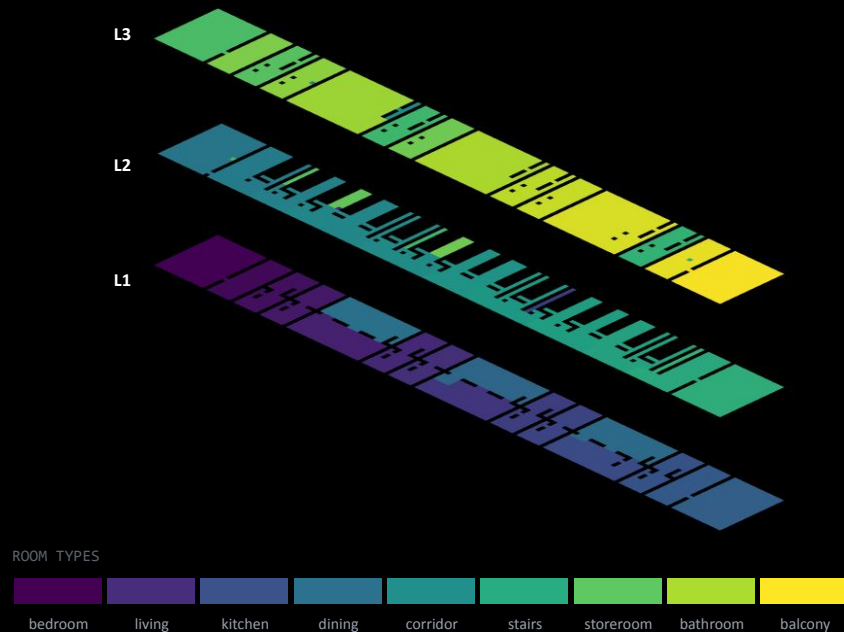
- Communities slice the building into longitudinal bands — each containing a corridor segment and its L2 rooms above.
- Western zone (purple/blue): kitchen block + west L2 rooms. Central zone (blue/teal): corridor core + stair-connected rooms.
- Community boundaries run perpendicular to the corridor — the building self-organises into vertical slices, not floor layers.
- Stairs bind each L2 room to its nearest L1 corridor segment more strongly than to other L2 rooms.

Community detection, Type F.

3-floor vertical slices
thicker bands than Type K

Same pattern extended
corridor binds rooms on both sides

Ginzburg's living cells
graph confirms integrated vertical units

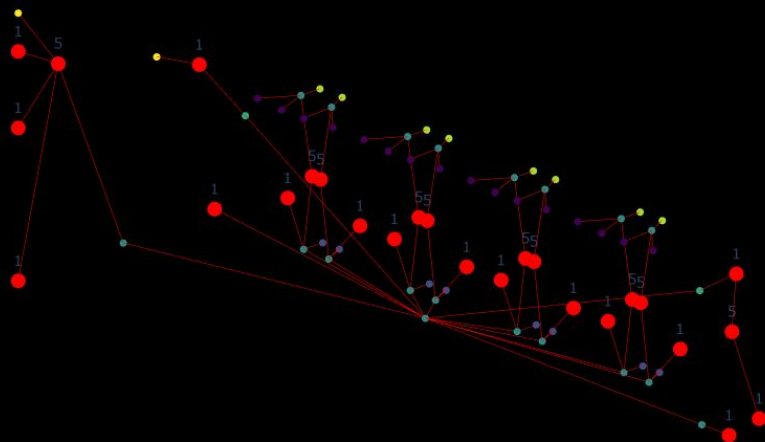


Key findings

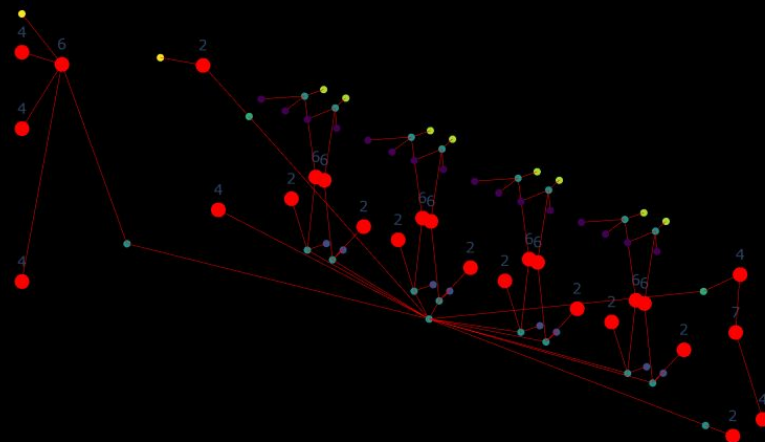
- Same longitudinal-band pattern, extended to three floors — each community spans a corridor segment plus L1 rooms below and L3 rooms above.
- Communities are thicker vertical slices than Type K — the middle corridor binds rooms on both sides.
- Community boundaries still run perpendicular to the corridor. Floors do not separate into their own communities.
- The building's true spatial units are three-floor vertical slices — consistent with Ginzburg's integrated living cells.

Room type prediction, K-type.

Ground truth



Prediction



NODES COLOUR ENCODES ROOM TYPE



Room type prediction, K-type.

67.9%

rooms correctly classified · 55 / 81

55

CORRECT

26

ERRORS

81

ROOMS

WHERE IT WENT WRONG · TRUE → PREDICTED

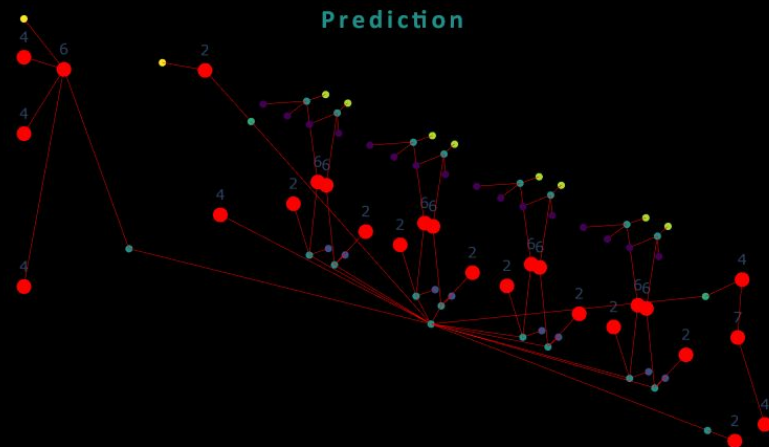
10 ■ Living room → ■ Kitchen

9 ■ Stairs → ■ Storeroom

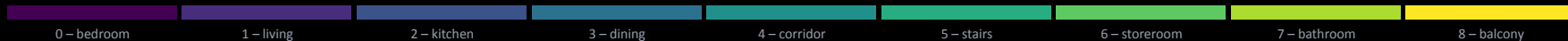
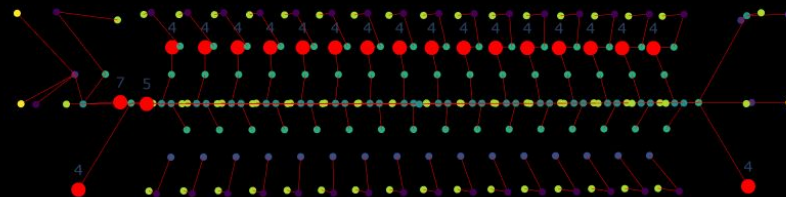
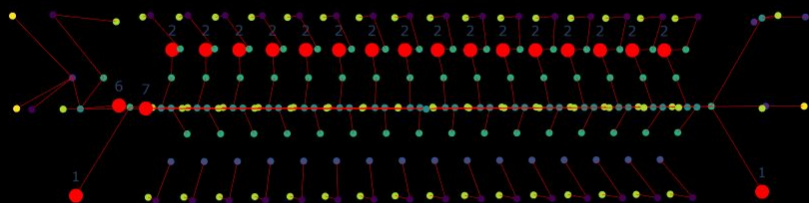
6 ■ Living room → ■ Corridor

1 ■ Stairs → ■ Bathroom

Errors cluster on rooms with near-identical connectivity, living rooms read as kitchens and stairs as storerooms.



Prediction



Room type prediction, F-type.

91.3%

rooms correctly classified · 210 / 230

210

CORRECT

20

ERRORS

230

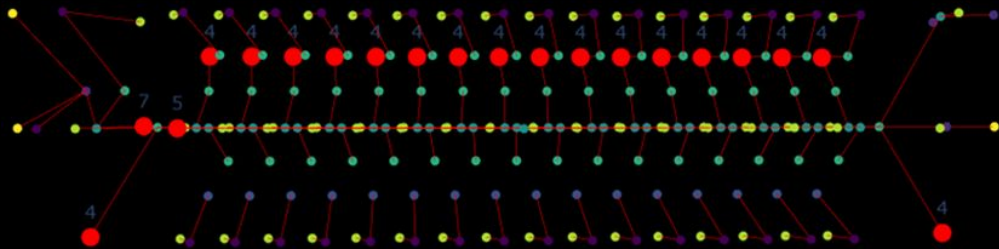
ROOMS

WHERE IT WENT WRONG · TRUE → PREDICTED

- 16 ■ Kitchen → ■ Corridor
- 2 ■ Living room → ■ Corridor
- 1 ■ Storeroom → ■ Bathroom
- 1 ■ Bathroom → ■ Stairs

16 of 20 errors are kitchens predicted as corridor, the single dominant blind spot; everything else is near-perfect.

Prediction



What the prediction tells us.

91.3%

F-type · 210 / 230 rooms

IN-DISTRIBUTION

vs

67.9%

K-type · 55 / 81 rooms

OUT-OF-DISTRIBUTION

Same pretrained model, same building — **two very different results.**

01

It generalizes — unevenly

F-type's duplex apartments resemble the Swiss Dwellings the model trained on, so it scores 91%. K-type's communal dom-kommuna layout is unlike anything in training. Accuracy falls to 68%.

02

Errors are systematic

Mistakes aren't random, rooms sharing a connectivity signature get swapped: living rooms ↔ kitchens, stairs ↔ storerooms, open kitchens read as corridor.

03

Topology captures convention

Connectivity alone reads conventional plans well. The Narkomfin's social-condenser program, its whole point, is exactly what challenges the model.